

Frozen but Adaptive

Cross-victim generalization of an RL-learned black-box adversarial attack policy

Research object The attack policy itself	Main setting Score-based, zero weight updates	Main test Held-out victim families
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Short answer: has anyone already done it?

Parts of it, yes. The exact study was not found in the reviewed literature. Researchers have used reinforcement learning to search attack augmentations, reweight surrogate ensembles, and train per-victim black-box attackers. We did not find an exact image-classification paper that trains one recurrent attack policy across source victim families, freezes all policy parameters, and evaluates zero-update behavior on completely held-out architecture families while clearly separating *policy transfer* from *adversarial-example transfer*.

What we should build

Train one recurrent RL policy on a population of source classifiers. Each episode samples a different victim, forcing the policy to learn attack behavior that is not tied to one model.

Freeze the policy, including its optimizer and parameters. Test it on victim architecture families never used for training, tuning, early stopping, or hyperparameter selection.

Use score-based black-box feedback as the main threat model. The frozen policy may observe query responses and adapt through recurrent hidden state, but it may not update its weights.

Compare against strong RL, query-based, transfer-based, and matched non-RL baselines under identical perturbation, query, data, and compute budgets.

Prepared as a decision-ready plan. Literature status is provisional and must be repeated immediately before submission. Date: 22 July 2026.

1. The Research Problem

The useful unit of transfer is the attacker, not only the perturbation it produced.

One-sentence objective

Determine whether a single frozen recurrent policy can use a small stream of black-box query responses to attack previously unseen classifier families more effectively and efficiently than existing reusable attack procedures.

What exists today

Most learned black-box attackers are trained or adapted for one victim. Transfer-attack research usually creates an adversarial image on a surrogate model and asks whether that same image fools another model. These are different objects: one transfers an **example**; the proposed work transfers a reusable **sequential decision policy**.

Why this is worth studying

Attack cost: per-victim RL training is expensive. A reusable policy amortizes training across many black boxes.

Realistic uncertainty: a deployed model's architecture and weights may be unknown. Security evaluation needs attacks that survive that shift.

Scientific value: the experiment isolates which attack behaviors are victim-invariant and which are model-specific.

Defensive value: a policy that generalizes across victims exposes systematic response patterns that defenses should address.

What would count as a contribution

Contribution	Required evidence
Method	A recurrent, architecture-independent attack policy trained with family-robust population objectives.
Protocol	A leakage-resistant leave-one-family-out benchmark that distinguishes example transfer, policy reuse, and target fine-tuning.
Result	Consistent gains in attack-success/query AUC or query efficiency on multiple held-out families, with uncertainty estimates.
Insight	Ablations and diagnostics explaining when and why cross-victim policy generalization succeeds or fails.

Not a breakthrough by default

The framing is promising, but novelty alone is insufficient. A conference paper needs a method that beats serious baselines, a clean held-out-family protocol, and an explanation of why RL is necessary. Without those, this is a benchmark observation or workshop paper, not a strong main-track result.

2. What Exactly Transfers?

Clear terminology prevents the core claim from collapsing during review.

Setting	Target information	Policy updates	Correct interpretation
T0: query-free transfer	No target feedback	None	Conventional adversarial-example or generator transfer.
T1: frozen score-based	Probabilities, scores, or top-k output	None	Main setting: cross-victim policy reuse with in-context adaptation.
T2: frozen decision-based	Top-1 label only	None	Harder secondary setting; tests whether minimal feedback is sufficient.
T3: limited adaptation	Target feedback	Small parameter update budget	Comparison only; measures the price and benefit of fine-tuning.

Main threat model: T1

The attacker receives a clean image and its true label. It may query a held-out classifier up to a fixed budget and observe scores or probabilities. It never receives gradients, weights, layers, training data, or architecture metadata. Policy parameters remain frozen. A recurrent state may change within an episode because that is inference-time memory, not learning.

Why not make zero-query the main claim?

Sequential RL is most defensible when actions react to environmental feedback. In a purely zero-query setting, reviewers can reasonably ask whether the method is only an attack generator that could be trained with supervised or evolutionary optimization. T0 remains an important endpoint, but T1 gives RL a clear technical role.

The conceptual boundary

TRAIN	FREEZE	DEPLOY	EVALUATE
Sample source victim families across episodes	Lock policy, encoder, normalization, and hyperparameters	Run on a never-seen victim using only allowed feedback	Measure success per query and across held-out families

Leakage rule

A held-out family must not influence training victims, validation victims, architecture choices, reward design, early stopping, or hyperparameters. If ViT is the test family, no ViT-derived signal may enter model selection.

3. Has Anyone Done It?

The honest answer is 'close work exists, but the exact protocol was not found.'

Work	What it already does	Why the proposed study is different
AutoMA [1]	Uses PPO to search model augmentation policies that improve transferable attacks.	RL searches augmentations; it is not a frozen direct perturbation policy evaluated across held-out victims.
L2T [2]	Dynamically selects transformations to improve transfer across CNNs, transformers, and APIs.	Transfers examples through transformations; no persistent zero-update black-box attack policy.
SMER [3]	Uses RL to reweight a surrogate ensemble for better transferability.	The policy controls surrogate weights, not direct sequential target attack actions.
RLAB [4]	Dueling DQN adds or removes perturbation patches for black-box attacks on several CNNs.	Reports per-victim attacks; no controlled frozen-policy transfer to held-out architecture families.
Adversarial Agents [5]	PPO learns a persistent black-box attack policy and generalizes to held-out images.	The reported generalization is primarily across images; adaptation to a new victim is left as future work.
QTRL [6]	DDQN-based query attack; reports cross-model success of generated adversarial examples.	The transferred object is the example, not a clearly frozen policy under a family-holdout protocol.
2026 benchmark [7]	Reviews and standardizes adversarial transferability methods, including RL-assisted ones.	Provides evaluation context, but does not establish the proposed policy-transfer claim.

Defensible novelty statement

After a targeted review of published and preprint literature through 22 July 2026, we did not find an exact image-classification study that trains one recurrent attack policy across source victim families, freezes all parameters, and evaluates its zero-update behavior on completely held-out architecture families while distinguishing policy transfer from example transfer. This is a provisional *to our knowledge* claim, not proof that no such work exists.

What cannot be claimed

- Do not claim the first use of RL for transferable adversarial attacks; AutoMA, L2T, and SMER are direct counterexamples.
- Do not claim the first persistent RL black-box attacker; Adversarial Agents is a close predecessor.
- Do not treat a perturbation made for model A that also fools model B as evidence that the policy generalized to B.

4. Research Questions and Claims

The paper needs one primary claim and several diagnostic questions.

ID	Question	Minimum evidence
RQ1	Can one frozen policy attack unseen architecture families?	Leave-one-family-out results across CNN, modern CNN, and transformer families.
RQ2	Does population training outperform single-source training?	Paired comparison using identical policy capacity, images, and environment steps.
RQ3	Does recurrent query context enable adaptation without fine-tuning?	Recurrent policy versus feed-forward and shuffled-history controls.
RQ4	Do calibration-independent observations improve cross-family behavior?	Ranks and score deltas versus raw logits/probabilities.
RQ5	Does RL add value over non-RL search or a matched generator?	Matched parameter/data/compute baselines plus Square Attack and SimBA-DCT.
RQ6	Which victim properties predict success?	Per-family diagnostics linked to calibration, gradients, decision margins, and architecture.

Primary hypothesis

At fixed perturbation and query budgets, a family-robust recurrent policy will improve the macro-average attack-success/query AUC on held-out classifier families over naive pooled PPO and the strongest comparable reusable non-RL baseline.

What makes the paper novel enough

Algorithmic novelty: recurrent victim-context inference plus worst-family training, not only pooled PPO.

Evaluation novelty: nested leave-one-family-out protocol with explicit zero-update verification.

Explanatory novelty: identify reusable attack primitives and failure boundaries across architecture families.

Success hierarchy

Outcome	Paper interpretation
Strong positive	Method plus benchmark paper: frozen policy wins across at least three held-out families.
Mixed but structured	Diagnostic paper: transfer is predictable from victim properties and feedback type.
Adaptation needed	Reframe around sample-efficient target adaptation; this requires a new algorithmic result.
Weak or noisy	Do not force a full paper. Release a careful benchmark or workshop report after replication.

5. Proposed Method

A victim-generalizable meta-attack policy with inference-time memory and no target-side learning.

1. OBSERVE	2. INFER CONTEXT	3. ACT	4. UPDATE STATE
Image state, class ranks, entropy, score change, query count	Recurrent hidden state estimates victim response behavior	Choose scale, region, direction, sign, and step size	Query target, receive allowed response, update hidden state only

Architecture-independent action space

Use multiscale image regions, DCT directions, or patch perturbation bases. Never use target gradients, layers, or feature maps.

Calibration-resistant observations

Prefer ranks, normalized score changes, entropy, success flags, remaining query fraction, and action history over raw logits alone.

Recurrent context inference

A recurrent PPO policy adapts its hidden state from query-response transitions. This is inference, not parameter fine-tuning.

Family-robust population training

Sample a source victim each episode and optimize a worst-family or GroupDRO-style objective so easy models do not dominate.

Strict projection and accounting

Project every step into the allowed norm ball and count every target call, including initialization and failed episodes.

Training objective

Start with recurrent PPO because the action sequence and partial observations form a natural partially observed decision process. Reward should combine attack progress, query cost, distortion cost, and success. Run a Dueling DQN pilot only if the action space is discrete enough to make the comparison meaningful.

Critical anti-shortcut test

Visualize action-location and frequency distributions across victims. If the learned policy merely repeats a fixed patch or frequency pattern, compare it against that deterministic strategy. A policy is only interesting if its decisions use query history or victim context.

6. Evaluation Design

The protocol is as important as the policy because leakage can manufacture apparent transfer.

Datasets and scale

Use CIFAR-10 and CIFAR-100 only for development, debugging, and broad ablations.

Use ImageNet-1K as the main benchmark. Stage with 1,000 correctly classified images, then confirm on about 5,000 images if compute permits.

Victim families

Family	Examples	Role
Classical CNN	VGG, ResNet, DenseNet	Source or held-out family
Modern CNN	EfficientNet, ConvNeXt	Source or held-out family
Transformer	ViT, DeiT	Source or held-out family
Hierarchical transformer	Swin	Source or held-out family
Robust models	At least two adversarially trained models	Stress-test distribution shift

Nested leave-one-family-out protocol

Outer loop: hold out one complete architecture family for the final test.

Inner loop: select hyperparameters only among source families, with a separate source-family validation split.

Repeat so each major family is held out. Report the macro-average across target models and target families.

Attack budgets

Dimension	Main values
Goal	Untargeted attack first; targeted attack only after the main protocol is stable.
Perturbation	L-infinity epsilon = 4/255 and 8/255; also report L2 and LPIPS.
Queries	0, 25, 100, and 500 target calls.
Seeds	At least five independent RL seeds.

Non-negotiable comparability

Do not collapse query-free, score-based, and decision-based attacks into one leaderboard. Compare methods only within the same information, query, perturbation, and compute budget, then present the three threat models as separate result blocks.

7. Baselines, Metrics, and Statistics

A credible result must survive both strong attacks and matched controls.

Group	Required baselines	Question answered
RL	RLAB-style DQN; Adversarial Agents-style PPO; per-victim; single-source frozen; naive pooled; target fine-tuned	Is the new training design better than known and simpler policy training?
Query-based	Square Attack; SimBA-DCT; Bandits or NES; HopSkipJump for T2	Does RL beat mature black-box search at the same query budget?
Matched non-RL	Feed-forward attack generator; random and greedy patch or frequency search	Is RL itself necessary, or only the action representation?
Transfer	MI-FGSM and DI-FGSM; L2T, SMER, and AutoMA where feasible	How does T0 compare with established adversarial-example transfer?

Primary and secondary metrics

Priority	Metric	Reporting rule
Primary	Attack-success/query AUC	Macro-average per target family; higher is better.
Primary	ASR at fixed budgets	Report at 25, 100, and 500 queries with 95% confidence intervals.
Secondary	Median queries to success	Treat failures as censored; do not discard unsuccessful attacks.
Secondary	Distortion and perceptual quality	L-infinity, L2, LPIPS, and valid-image rate.
Secondary	Runtime and training cost	Wall-clock, target calls, source training calls, and hardware.

Statistical design

- Use a hierarchical bootstrap over target family, target model, image, and RL seed to produce 95% confidence intervals.
- Use paired permutation tests on per-image outcomes when methods attack the same image-model pairs.
- Apply Holm correction across the primary family-level comparisons.
- Treat the target model or family, not every image independently, as the architectural generalization unit.

The baseline most likely to decide the paper

A matched feed-forward generator or fixed action strategy with the same parameter count, training data, source victims, and compute. If it performs as well as the recurrent policy, the paper cannot claim that RL or online context inference caused the gain.

8. Ablations and Research Deliverables

Each experiment should eliminate one plausible alternative explanation.

Ablation	Comparison	Interpretation
Victim diversity	Single source vs pooled sources vs worst-family training	Does diversity cause transfer, and does robust optimization help?
Memory	Recurrent vs feed-forward vs shuffled history	Is adaptation driven by temporal query context?
Observation	Raw scores vs ranks and normalized deltas	Is calibration shift a major barrier?
Action basis	Pixels vs patches vs DCT vs multiscale hybrid	Which attack primitives survive architecture shift?
Feedback	T0 vs T1 vs T2 vs T3	How much target information and adaptation are actually required?
Family holdout	Random model split vs strict family split	Does a conventional split overestimate generalization?
Reward	Success only vs query-aware vs distortion-aware	Which objective creates reusable behavior?

Required artifacts

A versioned victim registry with model family, checkpoint source, preprocessing, clean accuracy, and license.

A common attack API that records every target call and enforces norm projection centrally.

Configuration files that encode outer holdouts, inner validation, budgets, seeds, and evaluation images.

Raw per-image results, aggregate scripts, statistical tests, and plots reproducible from one command.

A model card for the attack policy documenting threat model, limitations, and responsible-use constraints.

Paper figures to design early

Figure	Purpose
Protocol diagram	Show source families, frozen boundary, held-out family, and allowed target feedback.
ASR-query curves	Primary result with one panel per held-out family and uncertainty bands.
Transfer matrix	Train-family rows by test-family columns to expose asymmetry.
Policy behavior	Action heat maps, frequency use, and hidden-state clustering by victim.
Failure map	Relate success to calibration, margin, robustness, and architecture.

9. Twenty-Four-Week Execution Plan

Run the smallest decisive experiment before investing in the full ImageNet benchmark.

Weeks	Work	Exit criterion
1-2	Lock terminology, threat models, literature table, victim-family split, metrics, and query accounting.	Protocol document is frozen before result collection.
3-5	Reproduce one PPO/DQN attacker plus Square Attack and SimBA-DCT in a shared harness.	Baselines match published trends and all queries are audited.
6-7	Single-source and naive pooled frozen-policy pilot on small held-out models.	Decide whether cross-victim signal exists.
8-12	Implement recurrent context, calibration-independent state, and worst-family training.	Method beats naive pooled policy on validation-family holdouts.
13-18	Run nested leave-one-family-out ImageNet experiments with five seeds.	Complete primary table and ASR-query curves.
19-21	Ablations, statistics, policy diagnostics, robust models, and failure analysis.	Every central claim has one falsification experiment.
22-24	Write, reproduce from clean environment, prepare artifact, and run internal skeptical review.	Paper, code, logs, and appendix agree exactly.

Week 7 go/no-go gate

Continue to full scale if	Stop or pivot if
The frozen policy beats random and greedy controls on several held-out models; population training improves over single-source training; seed intervals suggest a repeatable effect; decisions depend on history rather than a fixed pattern.	Gains vanish under matched query budgets; only seen architectures improve; a fixed perturbation strategy matches the policy; results reverse across seeds; or target-family tuning is needed to obtain the effect.

Practical sequencing

Do not begin with every model and dataset. First prove that the frozen-policy signal survives one CNN-to-transformer and one transformer-to-CNN holdout. Then scale the evaluation and add the full ablation matrix.

10. Review Risks and Publication Strategy

The strongest reviewers will attack the causal claim, not just the implementation.

Likely criticism	Required response
'RL transfer already exists.'	Acknowledge AutoMA, L2T, and SMER. Narrow the claim to frozen direct policy generalization under family holdout.
'This is example transfer.'	Evaluate the same frozen policy directly on each held-out victim and keep generated-example transfer in a separate T0 table.
'The target leaked into tuning.'	Use nested family holdouts, frozen configs, and a documented victim registry.
'RL is unnecessary.'	Include matched generator, fixed strategy, random search, greedy search, Square Attack, and SimBA-DCT.
'Query accounting is unfair.'	Centralize the target interface, log every call, and publish per-image traces.
'Results are toy-scale or unstable.'	Use ImageNet, heterogeneous victim families, robust models, five seeds, and hierarchical uncertainty.
'The novelty is only evaluation.'	Add recurrent context inference and family-robust training with evidence that each component matters.

Venue positioning

For a security venue, emphasize reusable black-box adversaries, realistic deployed-model uncertainty, query accounting, and defensive implications.

For an ML or vision venue, emphasize distribution shift over victim environments, recurrent context inference, family-robust training, and generalization analysis.

Choose the venue after the Week 18 result: security if the threat-model result is strongest; ML/vision if the algorithm and generalization result is strongest.

Responsible research practice

Use public benchmark models and authorized APIs only. Document rate limits and costs. Release the evaluation harness and aggregate analysis first; decide how to distribute pretrained attack policies after an institutional dual-use review. State limitations and intended defensive use clearly.

Recommended paper claim

We introduce and evaluate a family-robust recurrent black-box attack policy that is trained across source victims, frozen, and deployed without parameter updates to held-out architecture families. Our protocol separates policy reuse from example transfer and measures generalization under matched query and perturbation budgets.

11. Decision Checklist and References

Use this page to keep the project honest as results arrive.

Before calling the work publishable

1. The transferred object and threat model are named correctly in every section.
2. No held-out family influenced training, validation, early stopping, or hyperparameters.
3. All target calls are logged and counted, including failed runs and initialization.
4. The policy beats matched non-RL controls, not only weak RL baselines.
5. Results cover multiple held-out families, robust victims, and at least five RL seeds.
6. Confidence intervals and corrected paired tests support the primary claim.
7. Ablations show that memory, observation design, and robust population training each matter.
8. The literature search and novelty wording are repeated immediately before submission.

Key references

- [1] Y. Li et al. 'AutoMA: Towards Automatic Model Augmentation for Transferable Adversarial Attacks.' IEEE TMM, online 2021 / volume 2023. [Paper link](#)
- [2] Z. Zhu et al. 'Learning to Transform Dynamically for Better Adversarial Transferability.' CVPR 2024. [Paper link](#)
- [3] S. Tang et al. 'Ensemble Diversity Facilitates Adversarial Transferability.' CVPR 2024. [Paper link](#)
- [4] 'Reinforcement Learning Platform for Adversarial Black-box Attacks with Custom Distortion Filters.' AAAI 2025. [Paper link](#)
- [5] 'Adversarial Agents: Black-Box Evasion Attacks with Reinforcement Learning.' arXiv 2025; Findings of CVPR 2026. [Paper link](#)
- [6] 'Query-Efficient Two-Phase Reinforcement Learning Framework for Black-Box Adversarial Attacks.' Symmetry 2025. [Paper link](#)
- [7] 'Delving into Adversarial Transferability on Image Classification: Review, Benchmark, and Evaluation.' arXiv 2026. [Paper link](#)
- [8] 'Learning to Attack: Adversarial Transformation Networks.' AAAI 2018. [Paper link](#)

Bottom line

This is not an untouched area, but it remains a credible paper direction if the contribution is defined narrowly: zero-update cross-victim generalization of the attack policy, demonstrated with strict architecture-family holdouts and strong matched baselines. The first decisive task is the Week 6-7 pilot, not writing a large system before the signal is known.